

AJITH REDDY DEVARAPALLI

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Professional Summary:

Instrumentation & Controls Engineer with 6+ years of experience delivering automation, process control, and building management solutions within FDA-regulated pharmaceutical and biotechnology manufacturing environments. Proven expertise in Siemens Design CC/BMS, PLC, HMI, SCADA, and industrial control systems supporting HVAC, cleanroom, utility, and process operations. Experienced in commissioning, startup, I/O verification, loop testing, calibration support, process optimization, and troubleshooting of critical manufacturing equipment. Strong knowledge of process control strategies, instrumentation, P&IDs, electrical schematics, and GMP compliance to ensure reliable, validated, and efficient plant operations. Demonstrated success collaborating with QA, Validation, Engineering, Maintenance, Manufacturing, and external vendors to execute IQ/OQ/PQ, FAT/SAT, change controls, deviations

Technical Skills

- **Automation & Control Systems:** Programmed, configured, and troubleshot Siemens S7-1200/S7-1500 PLCs using TIA Portal for industrial automation and process control applications. Siemens BMS/Design CC, HVAC controls, utility systems, cleanroom environmental controls, alarm handling, control sequence troubleshooting, and process automation support
- **Process Automation & Manufacturing Support:** Process control loops, critical process parameters, equipment startup, manufacturing line support, process instrumentation, control logic optimization, sensor accuracy verification, and production troubleshooting
- **Commissioning & Qualification:** I/O checks, loop checks, functional testing, startup support, equipment turnover, FAT/SAT execution, IQ/OQ/PQ support, and system readiness testing
- **Pharma / GMP Compliance:** cGMP, GxP documentation, data integrity, deviations, CAPA support, change control, risk assessments, validation documentation, and audit/inspection support
- **Engineering Documentation:** P&IDs, wiring diagrams, control schematics, URS, FRS, test protocols, calibration procedures, validation reports, technical specifications, and maintenance documentation
- **Team Player-**Collaborated effectively with cross-functional teams including QA, Validation, Engineering, Maintenance, Manufacturing, Operations, and external vendors to deliver projects on schedule and in compliance with GMP requirements.

Professional Experience:

Novartis LLC, IN

Oct 2025 - Present

Instrumentation and controls Engineer

- Programmed, modified, and troubleshoot Siemens S7-1200/S7-1500 PLCs using TIA Portal to support HVAC, utility, and process automation systems.
- Developed and maintained HMI applications using Siemens WinCC, enabling real-time system monitoring, alarm management, diagnostics, and operator control.
- Working extensively with Siemens BMS (Desigo CC) for monitoring, alarm handling, and troubleshooting of HVAC and utility systems.
- Supporting reliable operation of building systems and utility infrastructure across facilities, with a focus on HVAC and cleanroom environments.
- Performing root cause analysis on equipment and system issues, implementing corrective actions to improve system reliability and reduce downtime.
- Supporting preventive and corrective maintenance activities for HVAC and associated utility systems.
- Assisting in installation, commissioning, and turnover of new equipment and instrumentation within facility systems, and Supported system qualification (IQ/OQ/PQ).
- Performing I/O checks, loop testing, and signal verification during commissioning and system modifications.
- Coordinate with vendors and contractors for system troubleshooting, maintenance activities, and small project execution.

Ford Motor Company, OH, USA

Automation Engineer

Jan 2024 - May 2025

- Supported PLC programming, commissioning, and validation of automated assembly and conveyor systems using Allen-Bradley ControlLogix/CompactLogix PLCs with Studio 5000 Logix Designer to ensure reliable production line operation.
- Conducted I/O verification, field device testing, and system checkout for conveyors, sensors, PowerFlex VFDs, safety circuits, and pneumatic equipment before production startup.
- Verified sequence logic, interlocks, and conveyor routing using Studio 5000 to ensure repeatable and fault-free operation across automotive assembly processes.
- Supported FactoryTalk View SE/ME HMI validation by verifying alarms, operator interfaces, system communication, and real-time production monitoring.
- Performed troubleshooting of PLC programs, FactoryTalk View, electrical controls, and EtherNet/IP networks to minimize downtime and improve equipment reliability.

- Collaborated with manufacturing, maintenance, electrical, and mechanical teams during installation, commissioning, production launches, and continuous improvement activities.
- Supported the integration and troubleshooting of industrial robots (ABB/FANUC) with PLC-controlled assembly and conveyor systems, verifying robot handshakes, safety interlocks, and production sequencing.
- Mentored junior controls technicians and electricians during commissioning and troubleshooting, providing guidance on Studio 5000, I/O validation, and system diagnostics to ensure consistent engineering practices.

Sandor Medicaids Pvt. Ltd.

Apr 2020 – Jul 2023

Service & Validation Engineer

- Developed FAT and SAT test scripts and participated in factory and site acceptance testing to ensure system readiness.
- Programmed, configured, and troubleshoot Siemens S7-1200/S7-1500 PLCs using TIA Portal for industrial automation and process control applications.
- Developed and modified PLC logic, function blocks, and control sequences using Siemens TIA Portal to meet operational requirements.
- Improved equipment uptime by ~40% by executing IQ/OQ protocols and applying structured troubleshooting to recurring system issues.
- Supported installation, commissioning (FAT/SAT), and operational readiness of diagnostic and medical systems across multiple client sites.
- Reduced recurring failures by performing preventive maintenance and root cause analysis, implementing CAPA-driven corrective actions.
- Ensured reliable equipment performance during critical procedures by supporting PQ validation under strict safety and compliance standards.
- Collaborated with QA, vendors, and cross-functional teams to resolve deviations and close CAPAs in compliance with QMS requirements.
- Maintained audit-ready documentation (GDP-compliant) for all service and validation activities.
- Delivered training to end users, reducing operational errors and improving compliance with maintenance procedure.

Education:

Master's in Biomedical Engineering.|Aug 2023- May 2025

Bachelor's in Electronics & Communication Engineering.|June 2017- Apr 2021